Department of Information Technology

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch : IV Semester-B. Tech Information Technology
Course Code & Title : IT3401 Web Essentials
Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic : Unit 5 / Servlet Life Cycle**
- **Course Outcome :** CO5
- **Topic Learning Outcome:** TLO43
- **Activity Chosen : Flipped Class Room**
- **Justification:**
  - Flipped Classroom is an innovative learner-centered teaching approach in which students study the fundamental concepts before attending the classroom session through learning materials such as videos, notes, and reference resources. Classroom time is then utilized for discussion, interaction, clarification, and collaborative learning activities.
  - This activity is adopted for the topic Servlet Life Cycle to encourage active participation, self-learning, and deeper conceptual understanding of servlet execution and lifecycle methods in web application development.

- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
  - Before the classroom session, students were provided with learning resources such as video lectures, concise notes, and presentation materials related to the Servlet Life Cycle.
  - Students were asked to explore the provided study materials on their own to understand servlet execution flow, lifecycle stages, and the purpose of lifecycle methods such as init(), service(), and destroy().
  - As part of the classroom activity, students were grouped into teams of two to four members and provided time to collaboratively discuss the concepts they had learned from the pre-class study materials, as shown in Fig. 1.
  - Following the discussion session, each group shared their understanding of the Servlet Life Cycle and described the functions of lifecycle methods in processing client requests within web applications, as illustrated in Fig. 2.
  - The instructor actively supported the discussion session by resolving students' misunderstandings, clarifying their doubts, and illustrating the real-time applications of servlet lifecycle methods through relevant examples.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO5	**2**	**2**	**3**	**2**	**2**	**3**

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	2	**2**	**3**	**2**	**2**	**3**
Justification for correlation	Students will be able to gain a strong understanding of the Servlet Life Cycle, servlet architecture, and lifecycle methods used in web applications.	Students will be able to understand and explain the functions and execution flow of servlet lifecycle methods effectively.	Students will be able to develop servlet-based applications and apply lifecycle concepts in practical web development scenarios.	Independent pre-class learning improves students confidence and self-learning ability in understanding servlet programming concepts.	Students will be able to improve their communication skills through group discussions, peer explanations, and presentations on technical concepts.	The servlet programming knowledge gained through this activity helps students apply web development concepts in real-time applications.

- **Images / Screenshot of the practice:**

Fig 1: Students discussion with their team members

Fig 2: Ram Selvalakshmi M and Kaviya K presentation

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students reported that the activity provided better clarity on the stages and working process of the Servlet Life Cycle.
    - Many students shared that prior preparation helped them to understand the servlet lifecycle methods more clearly during the interactive session.
    - The activity helped students clearly understand the role and practical significance of servlet lifecycle methods in web applications.
  - ***Benefit of the practice:***
    - Through the activity, students gained clear knowledge about the Servlet Life Cycle and the purpose of methods like init(),service(),and destroy().
    - The availability of pre-class learning materials allowed slow learners to revisit concepts multiple times for better understanding.
    - Students gained confidence in explaining servlet concepts and understanding the execution process of servlets in web applications.
  - ***Challenges faced in implementation:***
    - Some students found it difficult to understand the execution flow of servlet lifecycle methods during the initial discussion session.
    - Limited classroom time made it challenging for every group to present their understanding in detail.
    - Technical terms related to servlet execution and lifecycle management were initially difficult for some students to understand.

In the subsequent class, further clarification and practical examples on the Servlet Life Cycle were discussed to reinforce students understanding of servlet lifecycle methods.

References:

- https://www.ritrjpm.ac.in/images/B.Tech-IT/20232024/IP/4_CS3352_VA_LT.pdf
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/MVB_SS_CS3492_Flipped_Classroom.pdf

Signature of Faculty Member **HOD**